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First Application of Autonomous Inflow Control Valve (AICV) in Libya to Reduce Water Production and Increase Oil Production

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Abstract

In line with a growing industry trend and as more wells are completed and retrofitted with inflow control technologies, the impact on production efficiency is becoming increasingly evident. This paper presents a case study on operational modeling, deployment, and well operations best practices workflow.

A mature brownfield developed with vertical wells, most of which experienced significant water production (for example high water cut), resulted in uneconomical and inefficient recovery. To mitigate the risk of water production and crossflow issues in an existing well, an operator performed a re-entry and completed the well using an Autonomous Inflow Control Valve (AICV).

Well B, a deviated observation well with 988 feet of open-hole length in a sandstone reservoir, had been classified as non-producing for 17 years. The operator sought to convert this well into a production well. However, due to the high water cut in nearby offset wells (approximately 71-74%), production was not feasible without implementing an AICV downhole completion to reduce water production and maximize oil recovery.

In March 2024, the AICV downhole completion was successfully installed to mitigate the anticipated water production and enhance oil production. Lower completion with AICV effectively shuts off water-producing zones and stimulates oil-bearing zones, significantly increasing oil output. Post-installation results show a water cut of around 3% and oil production of approximately 1,134 bbl/d. Compared to offset wells, water production was reduced by 98% from 1,632 bbl/d to 34 bbl/d. Additionally, water cut dropped from 71% to 3%, while oil production increased by 72%, from 661 bbl/d to 1,134 bbl/d.

In line with a growing industry trend and as more wells are completed and retrofitted with inflow control technologies, the impact on production efficiency is becoming ever more evident. This paper outlines key learnings for activities prior to, during, and post deployment of AICV lower completion as applied to this well; it could serve as a guideline highlighting key considerations of AICV understanding and modelling for future AICV applications.

Well Background and Reservoir/Production challenges

The reservoir intervals along well trajectory are sandstone, ranging from fine to medium and medium to coarse grain sizes. Mineralogical analysis indicates the presence of kaolinitic clays, along with iron-bearing minerals such as pyrite and hematite, within the reservoir sands. These minerals may impact reservoir quality, particularly permeability and fluid flow behavior. Net Oil Pay relies on the current Oil-Water Contact (OWC) interpretations derived from nearby wells. Utilizing the highest OWC observed in re-entry Well C, a vertical Net Oil Pay thickness of approximately 45 feet has been identified across the two reservoir layers.

Well B was originally completed as an oil producer in late 1993. However, it remained non-productive until mid-2001, at which point it began producing at a low rate. During its operational life, an Electrical Submersible Pump (ESP) was installed to improve production, but no sustained flow to surface was achieved. The well was acidized and swabbed, recovering 100% water. Subsequently, well was shut in November 2004: all perforated intervals were squeezed with cement. A re-perforation was conducted afterward, and swabbing operations recovered 100% oil. However, due to poor reservoir connectivity and low fluid entry, the well was suspended pending further evaluation. In a 2007 workover campaign, additional perforations were made in an effort to improve reservoir inflow and restore production. Despite these efforts, the well continued to show very low influx and a deep fluid level, with production consisting entirely of water. Given these results, it was ultimately recommended that Well B be converted into an observation well for future reservoir monitoring.

A re-entry sidetrack is proposed to optimize oil recovery from two target reservoir layers located in the central part of the field. The planned well trajectory is designed to access two reservoir layers (fault block and second block segments) between offset wells J and Q, thereby improving drainage efficiency and reservoir sweep. Relative to the surrounding structural setting, the re-entry target lies about 20 feet deeper than one neighboring fault block and approximately 40 feet deeper than another adjacent block. This structural positioning has implications for fluid contact levels and reservoir connectivity.

Production test data from offset wells indicate mixed reservoir performance. Tests conducted in 2013 and 2023 show water cut (WC) levels exceeding 70%, with oil production rates ranging between 358 and 661 stbpd. These high-water cuts suggest potential challenges with reservoir heterogeneity, water breakthroughs, or coning in the area.

Table 1—Offset well test production

Well Name	Oil Rate, stb/d	Water cut, %
Well J	478	74
Well Q	358	74
Well V	661	71

Despite the elevated water production, the oil rates confirm that the reservoir remains productive under the right conditions. This highlights the importance of careful placement and reservoir management of well. This proposed re-entry horizontal location is designed to develop and more effectively drain oil in two reservoir layers in an uplifted fault block located between offset oil producers Well J and Well Q.

By drilling in this structurally advantageous position, the re-entry well aims to more effectively drain remaining oil, enhance reservoir contact, and delay water encroachment—ultimately improving field recovery and extending productive life.

Principle of AICV Technology

The Autonomous Inflow Control Valve (AICV) technology developed to delay the onset of water and/or gas breakthrough, and in case of breakthrough, it will restrict the production of these unwanted fluids such as water, gas, and steam significantly. The AICV distinguishes between reservoir fluids and reacts accordingly.

The AICV design is truly autonomous and is based on the Hagen-Poiseuille's equation and Bernoulli's principles. The functionality is based on the viscosity and density of the reservoir fluids. The AICV is to open for high viscous fluid and close for low viscosity fluid. Which means the valve keeps open for oil and closes for lower viscous fluids such as water and gas. Fig. 1 describes the elements of AICV along with various forces acting on the piston. Each AICV consists of the main flow from where reservoir fluid enters the device and a pilot flow which represents 3-5% of the total flow going through a device. The pilot flow consists of a Laminar Flow Element (LFE) and the Turbulent Flow Element (TFE) with a pressure chamber located between them.

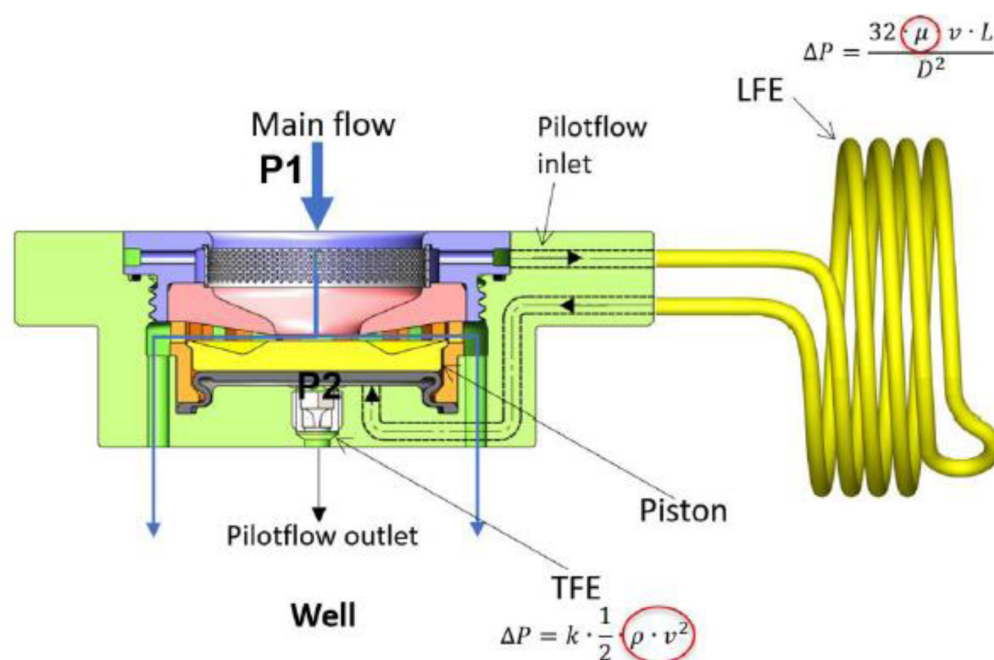


Figure 1—AICV Design.

As it is illustrated in Fig. 1, the reservoir fluid at reservoir pressure P1 enters the valve inlet, and a minor portion of the fluid is guided through the pilot flow elements, while the main flow passes through the small opening between the inlet seat and the piston. By passing through the LFE, the pressure of the fluid is reduced to P2 pressure at a chamber called the P2 chamber. The fluid properties and the flow rate through the pilot flow determine the P2 pressure. The pilot flow is further led through the turbulent flow element in which the second pressure drop to the well pressure, P3, occurs. The P2 pressure in the P2 chamber controls the valve functionality. Different P2 pressures act on the piston based on the different fluids passing through the LFE. For higher viscosity fluids like oil, the pressure drop through the LFE is high, resulting in a lower P2. Because the P2 pressure is low for oil, the generated force under the piston, which acts upward, is low as well. This will keep the piston in an open position, producing oil to the well through both the main flow and the pilot flow path. For lower viscosity fluids, such as gas and water, the pressure drop through the LFE is low, resulting in a higher P2. Because the P2 pressure is high for gas and water, the generated force under the piston, which acts upward, is high as well. This will actuate the piston and the valve will close. The closed valve produces gas and water to the well only through the pilot flow path.

The AICV is designed to work with various oil viscosity ranges—from ultra-light oil to extra-heavy oil. Extensively single-phase and multi-phase flow loop tests have been performed to verify and qualify the AICV. The experimental results are also used to create empirical equations and coefficients to model the AICV flow performance. These equations and relevant coefficients are used to perform simulation in many commercial near-well and numerical reservoir simulators to simulate AICV completions and to evaluate the benefits of it in specific reservoir conditions and completion architectures.

Fig.2 shows typical AICV screen assembly along with a packer for zonal isolation: AICV mounted in a base pipe with sand screen, where AICV is covered by housing. Each compartment can have one or more AICVs. Compared to a traditional sand screen, the AICV joint base pipe is unperforated under the screen section. The reservoir fluids enter into the screen and then flow underneath the screen to enter into AICV housing. Furthermore, the fluid flows through AICV and enters into the production tubing. As the fluid passes through the AICV, a pressure drop occurs across it as a function of fluid properties and AICV geometry.

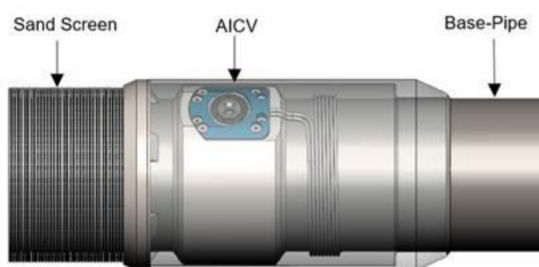


Figure 2—AICV mounted in a base pipe with sand screen.

Annular swellable packers must be in place to provide robust zonal and annulus isolation. Compartmentalization is one of the most crucial factors in the optimization of completion design with Autonomous Inflow Control Valve. It directly influences performance and recovery efficiency. The effectiveness of compartmentalization is largely determined by key reservoir characteristics, including permeability variations, fluid saturation distributions, and overall reservoir information.

Operational modeling. Optimizing well design

Well performance flow modeling serves as a key component in assessing the feasibility and effectiveness of an Autonomous Inflow Control Valve application. Given the inherent uncertainties in subsurface data, modeling serves as a valuable tool to analyze various reservoir conditions and predict the potential impact of an AICV completion on overall performance. By simulating different production scenarios, modeling allows operators to identify potential risks and optimize design well. Furthermore, it provides insights into fluid flow dynamics, pressure distribution, and production rates, ensuring that the AICV completion system is appropriately configured for maximum efficiency. In addition to evaluating performance outcomes, well modeling aids in determining the optimal quantity and placement of AICV completion equipment and zonal isolation packers, helping to minimize costs and improve long-term operational success.

Lower completion well simulation is generally performed using a near-wellbore (NWB) completion hydraulics steady-state simulator. Operational modeling is conducted to further optimize the well design to meet production targets. The AICV completion design optimization process begins once the well is drilled and actual well data is received. After the well design is optimized, a run-in-hole tally is prepared to match the final design, and the completion string is deployed and installed in the well.

Selecting the optimal well completion, including the quantity and placement of isolation packers and AICVs along the wellbore, is crucial for minimizing water production and optimizing net oil production and ultimate recovery.

The key stages of the AICV simulation and completion design and optimization process workflow completion design are illustrated in Fig. 3.

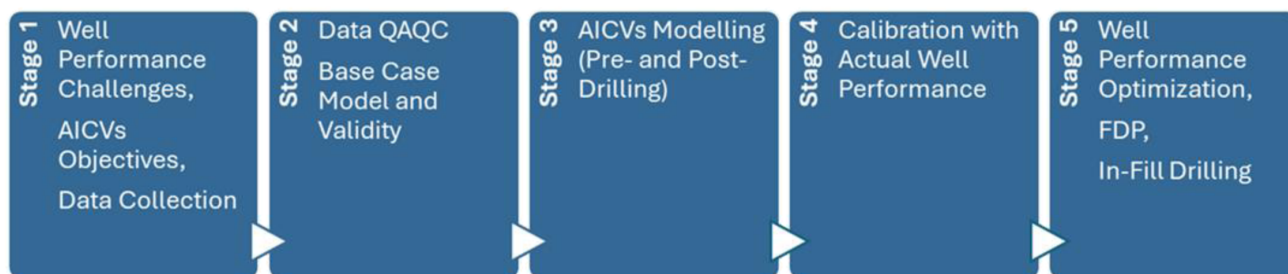


Figure 3—AICV completion design and optimization workflow.

The workflow begins at Stage 1 with a discussion with an operator's multi-disciplinary team about an overview of a reservoir, field development strategy and well performance challenges. AICV completion objectives and KPIs along with the well production strategy are discussed at this stage. Also, at this stage, reservoir, fluid, well completion and production (offset/model forecast) data etc. are collected. In Stage 2, a base case model of existing legacy completion such as stand-alone screens or slotted liners or bare-foot completion is built. Based on the evaluation, this base-case model is completed with a sensitivity analysis such as variation in permeability and saturation along the production interval, location of fractures/faults, decline in reservoir pressure etc. to evaluate the conventional completion well flow performance during the life of the well. As the base case model serves as a baseline to compare with well flow performance of the AICV completion, data QAQC is performed, and necessary adjustments are made.

Stage 3 is AICV modelling before actual well drilling, the AICV completion model is set up by defining the zonal isolation packers for compartmentalization and AICV joints in each segment. The model is run and well inflow performance, reduction in unwanted fluid, and gain in oil production are analyzed. Once the candidate well is drilled, the pre-drilling model is updated with the actual survey, permeability, saturation as well as other well logs. This post-drilling model is adjusted with the number and depth of zonal isolation packers and the number of AICVs per compartment. The model is subjected to various sensitivities and results are analyzed with an operator's team to finalize the post-drilling AICV completion design.

In Stage 4, after a sufficient clean-up period and for the stabilized well production, the post-drilling AICV model is calibrated by adjusting the required parameters such as pressure, permeability, saturation etc. in the model. This calibration, especially for the first few well test data, is very critical to address anomalies/uncertainty within reservoir data, errors in well log interpretations, unknown or inaccurate subsurface properties, unidentified faults or high permeability zones, human error in log interpretation, data resolution etc.

The final stage of the completion design and optimization workflow includes operational modeling and well design optimization, which involve the following steps:

- Data Collection and Storage: Actual well data from the operator is received and reviewed.
- Model Update and Sensitivity Analysis: The model is updated with the new data, and required sensitivities or modifications are re-run based on the completion meeting requirements and downhole completion objectives.
- Report Preparation: A final AICV completion design optimization report is prepared and shared.
- Report Distribution and Implementation: The approved AICV completion design optimization report is shared with the operator and the operational team to create a downhole AICV completion schematic/tally.

Open hole isolation packers are used to create compartments in AICV completions to ensure that the formation with different reservoir attributes is isolated. Because pressure-drop requirements in each compartment vary according to near-wellbore geological conditions, compartmentalization is vital to avoiding the annular flow and for the efficient functioning of AICV technology. Without effective compartmentalization, there will be an annular flow, due to which there will be a uniform pressure drop across the entire well.

Additionally, factors such as shale content, the presence of natural or induced fractures, and heterogeneities within the formation play a significant role in planning the compartmentalization strategy. Properly assessing these parameters ensures that the AICV system is designed to effectively manage inflow control, minimize water or gas breakthroughs, and enhance hydrocarbon production. By carefully analyzing and optimizing compartmentalization, operators can achieve improved sweep efficiency, prolong the productive life of the well, and maximize overall reservoir recovery.

AICV Well Performance

A steady-state hydraulic simulator was used to model fluid flow through the reservoir and AICV downhole completion on Well B. The simulator predicts reservoir inflow based on near-wellbore reservoir properties. Using the provided data and the completion of the Vistrack inversion, the formation above and the target formation were separated. AICV compartments were employed to differentiate contrasts in reservoir rock and fluid properties, typically identified in open-hole logs or petrophysical reservoir description interpretation logs. AICV redistributes downhole pressure based on the pressure drop across each compartment. Due to the lack of detailed information for well B, the section along the target formation was equally divided into nine segments (Fig.4). Segmentation involves dividing the open-hole length into smaller compartments to optimize reservoir management.

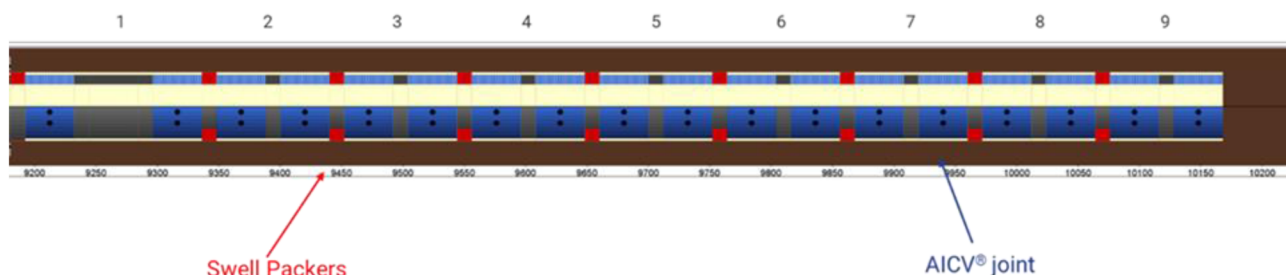


Figure 4—Well completion with AICVs.

The operation model used average parameters value such as saturation and permeability along the wellbore from the offset wells information. One of the critical points was to segregate the first zone from the heel which was along the first reservoir layer (fault block), where rest of the compartments produced from the second block segment.

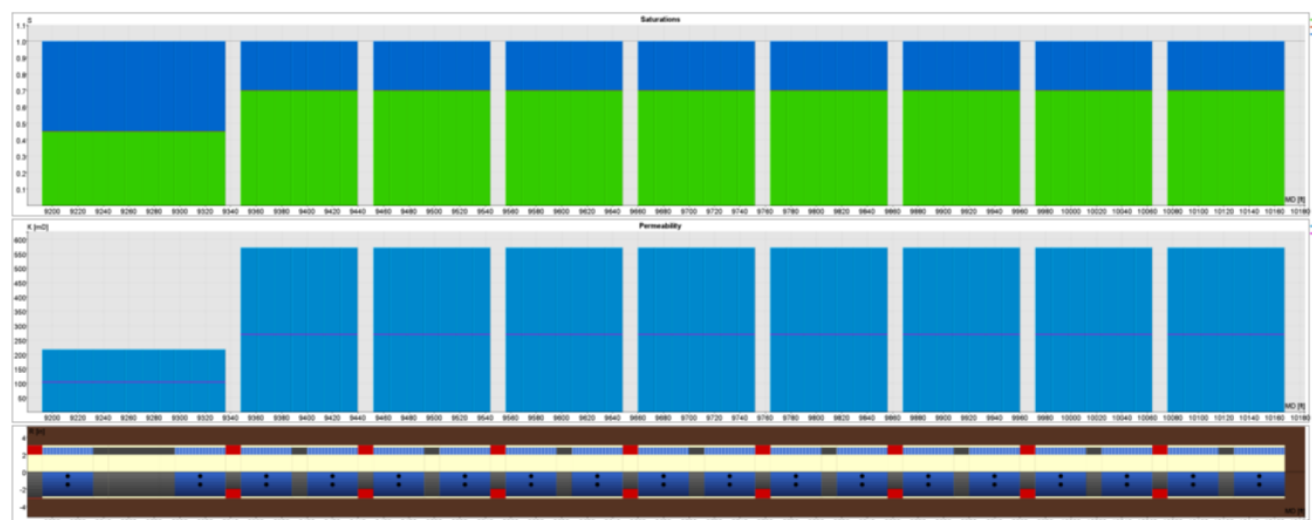


Figure 5—Formation properties along Well B.

At the time of this publication, one well has been installed with AICV completion in the field. The AICV completion proved not only a significant reduction in un-wanted water production in these matured fields but also enhanced the oil production from the wells. Thus, increasing the recovery from the well.

Result

In March 2024, the AICV down hole completion was successfully installed in this well to reduce the expected high-water production and increase the oil production. The open hole section was divided into 9 compartments equipped with 18 dual AICV joints (2 AICV's per joint). The AICV was able to stop the water production zones/compartments and simultaneously stimulate the oil-bearing zones to produce more oil.

For the first time in North Africa, the latest Autonomous Inflow Control Valve (AICV) technology from Norway, was successfully used to reduce associated water. The operator of the field was able to convert the non-producing Well B, which was classified as an observation well for 17 years, into a productive well by directed and re-entry drilling and using modern technologies.

The results after the AICV installation, shows water cut around 3% and oil production around 1134 bbl/d. By comparing this result with offset wells, the water production is reduced by around 98% from 1632 bbl/d to 34 bbl/d, water cut is dropped from around 71% to 3%, and oil production has increased by 72%, from 661 bbl/d to 1134 bbl/d. which shows that the flow influx along the horizontal well can be improved by using AICVs.

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